

CE 35000: FLUID MECHANICS PROJECT
REPORT

Simulation and Analysis of Channel Flow With High-Temperature Injection Using ANSYS Fluent

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Contents

Acknowledgements	i
Contents	ii
Abstract	1
1 Introduction	2
2 Flow Simulation Using ANSYS Fluent	3
Model Setup	3
Flow Visualization	3
Pathlines and Streamline Vectors	4
Velocity Contours	6
Vorticity Contours	7
Pressure Contours	8
3 Analysis of the Flow	9
Application of Fundamental Principles	9
Continuity Analysis Across Cross-Sections	10
Momentum Analysis Using Pressure Distribution	11
4 Conclusion	13
References	14
Appendices	14

List of Figures

Figure 1	Case A Pathlines.	4
Figure 2	Case A Streamline vectors.	4
Figure 3	Case B Pathlines	5
Figure 4	Case B Streamline Vectors	5
Figure 5	Case A Velocity Contours	6
Figure 6	Case B Velocity Contours	6
Figure 7	Case A Vorticity Contours	7
Figure 8	Case B Vorticity Contours	7
Figure 9	Case A Pressure Contours.	8
Figure 10	Case B Pressure Contours	8
Figure 11	Case A Scaled Residuals	15
Figure 12	Case B Scaled Residuals	15
Figure 13	Case A Drag Coefficient	15
Figure 14	Case B Drag Coefficient	16
Figure 15	Mass flow report for cases A and B.	16
Figure 16	Force report for case A.	16
Figure 17	Force report for case B.	16

List of Tables

Table 1	Inlet velocity settings for cases A and B.	3
Table 2	Mass flow rates at all boundaries for cases A and B.	10
Table 3	Comparison of momentum balance for cases A and B.	11

Abstract

Different inlet and injection velocity conditions were explored to examine the behavior of discharge flow within a channel using ANSYS Fluent. This project aims to provide scientific reasoning for how flow patterns, pressure fields, and velocity distributions develop in channel configurations, and to understand the relationships between fundamental fluid mechanics principles and the simulated results. By evaluating streamline patterns, velocity contours, and pressure variations for both low-velocity and moderate-velocity cases, the study seeks to determine how changes in boundary conditions influence the overall flow behavior.

Core concepts such as mass conservation and momentum balance are used to quantify flow rates across selected cross-sections and to assess the accuracy of the simulated results. It was found that higher inlet and injection velocities produce more defined streamline curvature, greater mixing near the injection region, and more pronounced pressure gradients along the channel. Mass conservation and momentum balance checks were also performed to verify the numerical accuracy of the simulations and ensure consistency with the governing fluid mechanics equations. These findings demonstrate how computational fluid dynamics can be used to model realistic flow scenarios and validate theoretical principles taught in fluid mechanics.

1 Introduction

Understanding how fluids behave under different boundary conditions is essential in describing both natural and engineered systems. In this project, the flow within a rectangular channel is examined, where a primary inlet stream interacts with an orthogonal injection stream. The purpose of this study is to analyze how these two flows interact, how velocity distributions develop throughout the channel, and how pressure variations arise due to the presence of an injection jet.

The channel is modeled using ANSYS Fluent, a computational fluid dynamics software that allows the governing equations of fluid motion to be solved numerically[2]. By simulating two different velocity conditions: one in which the imposed flows are extremely small, and a second in which both the inlet and injection velocities are larger, the project aims to identify how these changes influence the resulting flow field. This includes observing the formation of streamlines, analyzing pressure contours, and assessing how momentum is distributed throughout the channel.

Flow phenomena such as mixing, jet penetration, and pressure gradients are expected to vary significantly depending on the magnitude of the velocity. These behaviors are governed by fundamental fluid mechanics principles, including the continuity equation, momentum conservation, and pressure-velocity relationships. Applying these principles to the simulation results will provide a clearer insight into the development of flow within the channel.

Overall, the objective of this project is to create a numerical model that reveals the structure of the flow inside the channel and to interpret these results using the concepts learned in CE 35000: Fluid Mechanics. Such analysis provides scientific reasoning for why the flow behaves the way it does under different conditions and establishes clear relationships between the imposed velocities and the resulting flow properties.

2 Flow Simulation Using ANSYS Fluent

Model Setup

Following the instructions provided, the rectangular channel geometry was imported into ANSYS Fluent, with water defined as the fluid. The boundary conditions included:

Case	Main Inlet Velocity (m/s)	Injection Inlet Velocity (m/s)
A	0.0001	0.0003
B	0.2	0.4

Table 1: Inlet velocity settings for cases A and B.

The mesh was generated using default settings, and the simulation was run until residuals reached acceptable convergence. Data was extracted along multiple cross-sections and surfaces, following the report guidelines in the project document.

Flow Visualization

To visualize how the flow develops within the channel, several post-processing tools within ANSYS Fluent were used to generate streamlines, velocity magnitude contours, and pressure contours for both velocity conditions. After the simulation reached convergence, data was extracted along selected cross-sections and surfaces of the channel. The streamline plots were used to observe the interaction between the inlet flow and the injection jet, while the velocity magnitude and pressure contour plots provided a clear representation of how these quantities vary throughout the domain. All figures were exported at high resolution using Fluent's graphics window to ensure clarity and accuracy, and each visualization was examined to identify key flow features such as mixing behavior, flow acceleration, and regions of pressure variation.

Pathlines and Streamline Vectors

Pathlines and streamline vectors were generated to visualize how the fluid moves throughout the channel and how the injection jet interacts with the main flow. These plots help illustrate changes in flow behavior between the low-velocity case (Case A) and the higher-velocity case (Case B).

Case A displays smooth, steady streamlines that remain close to horizontal across the channel. Since the velocities in Case A are very small, the flow remains laminar and moves smoothly through the channel with only mild variations near the injection region, producing slight curvature in the immediate vicinity of the circular inlet. No recirculation zones or unsteady behavior are observed. The streamline vectors show uniform flow with very small magnitudes, consistent with the low Reynolds number of this case.

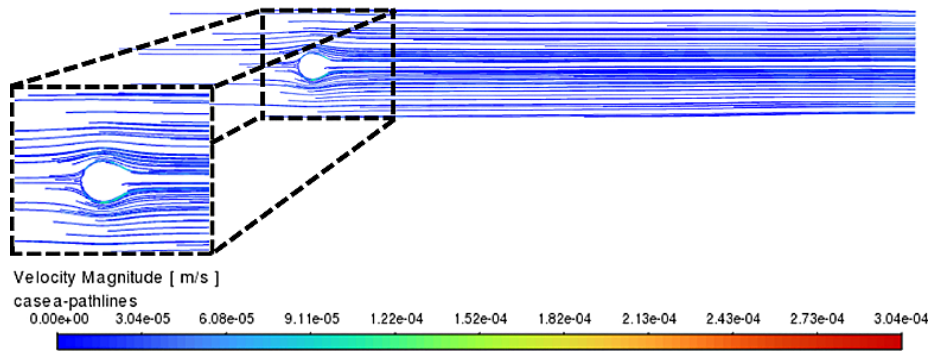


Figure 1: Case A Pathlines.

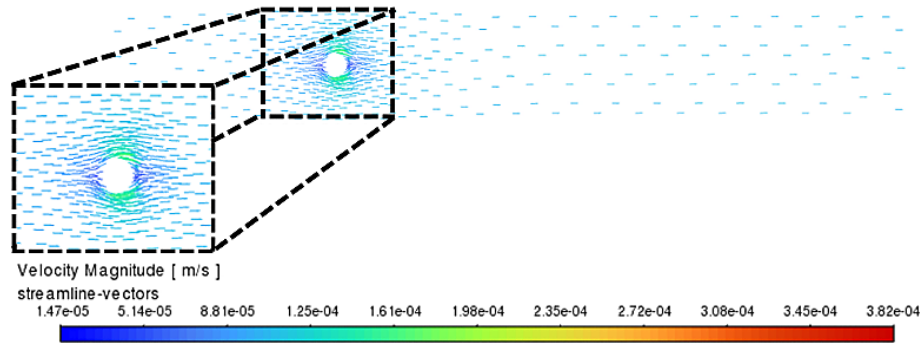


Figure 2: Case A Streamline vectors.

In Case B, the pathlines and vectors clearly show a stronger jet issuing from the circular opening, resulting in noticeable acceleration and curvature of the flow. The injection stream goes further into the main channel and drives a more pronounced mixing layer. Downstream of the injection point, the pathlines diverge and spread as the higher-momentum jet draws in the surrounding fluid. Although the flow remains steady in this simulation, the streamline vectors show a sharper velocity gradient and a more developed downstream flow field compared to Case A. These differences arise from the significantly larger inlet and injection velocities used in Case B, which increase momentum transport and strengthen the overall flow structure.

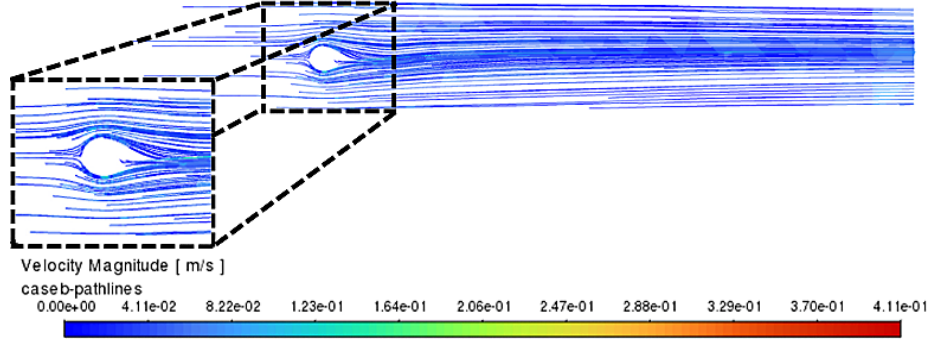


Figure 3: Case B Pathlines

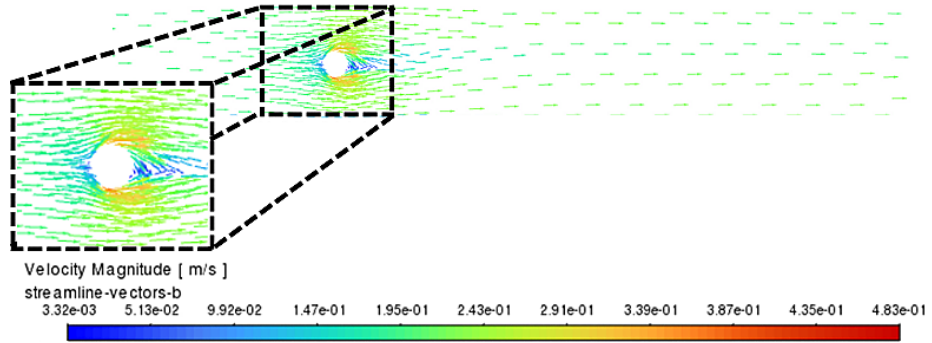


Figure 4: Case B Streamline Vectors

Overall, the pathlines and streamline vectors illustrate how increasing inlet and injection velocities produce more energetic, curved, and mixed flow patterns, while very low velocities result in smooth motion with minimal disturbance.

Velocity Contours

Velocity contours were generated to visualize how the flow acceleration changes around the injection region.

In Case A, with very small inlet velocities, the flow shows only mild velocity variation. The injection jet slightly disturbs the main flow in the channel, but no strong acceleration zone forms because the Reynolds number is extremely low. The flow remains smooth, symmetric, and nearly uniform across the channel.

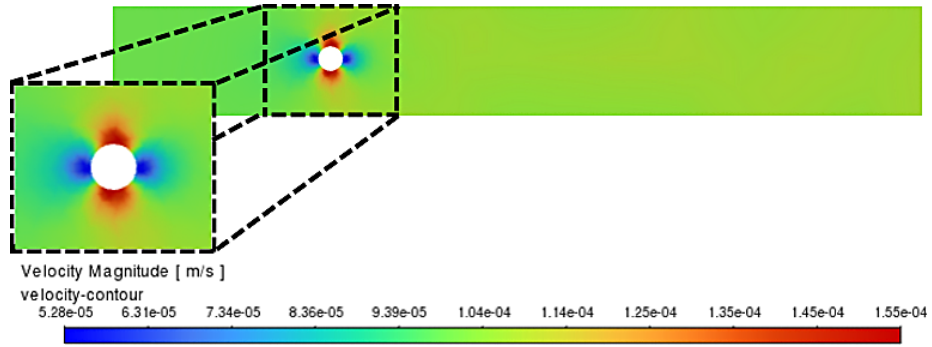


Figure 5: Case A Velocity Contours

In Case B, with higher velocities, shows a defined jet issuing from the circular injection region. The jet accelerates the surrounding fluid, creating a visible shear layer that spreads downstream. This leads to stronger mixing and a larger region of elevated velocity magnitude. These effects are expected because increasing the inlet and injection velocities increases momentum and the instability of the mixing layer.

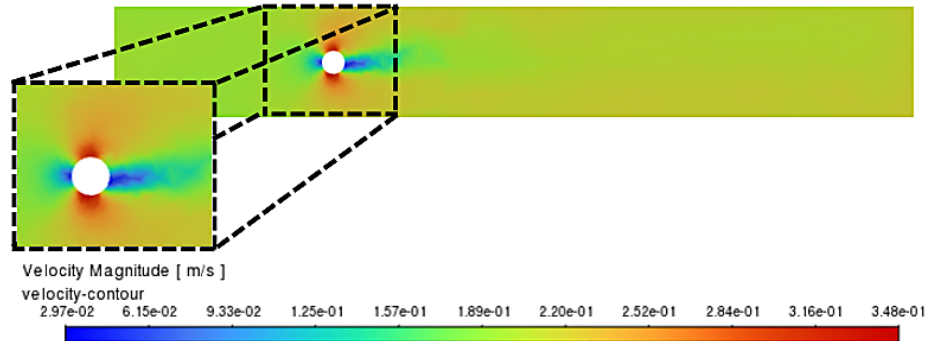


Figure 6: Case B Velocity Contours

Vorticity Contours

Vorticity contours were used to examine rotational behavior in the flow.

In Case A, the vorticity contours show localized rotation near the injection region. This occurs because even at low velocity, vorticity is generated by shear, not speed alone. Since the injection creates a strong velocity gradient between the jet and the surrounding fluid, the region around the inlet generates measurable rotational effects despite the low speeds.

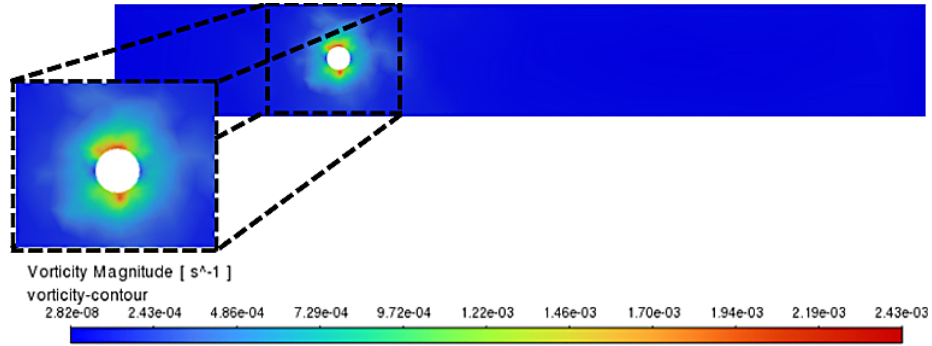


Figure 7: Case A Vorticity Contours

In Case B, the vorticity field is more intense near the injection region. Compared to Case A, the rotational region is more spread out, which is consistent with a higher-momentum jet that penetrates further into the main flow in the channel. Visually, the vorticity appears similar to Case A. This is because Fluent automatically scales the contour range[2]. When using the same scale for both cases, the Case B vorticity is significantly larger.

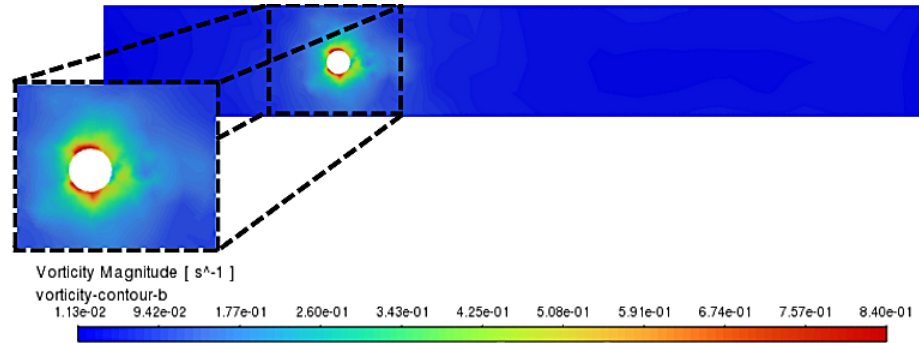


Figure 8: Case B Vorticity Contours

Pressure Contours

Pressure contours illustrate how pressure decreases along the flow direction and how the injection region affects the field.

In Case A, the pressure disturbance is minimal because velocities are extremely small. Only a slight low-pressure region forms around the injection jet, and the pressure field remains nearly uniform elsewhere in the channel.

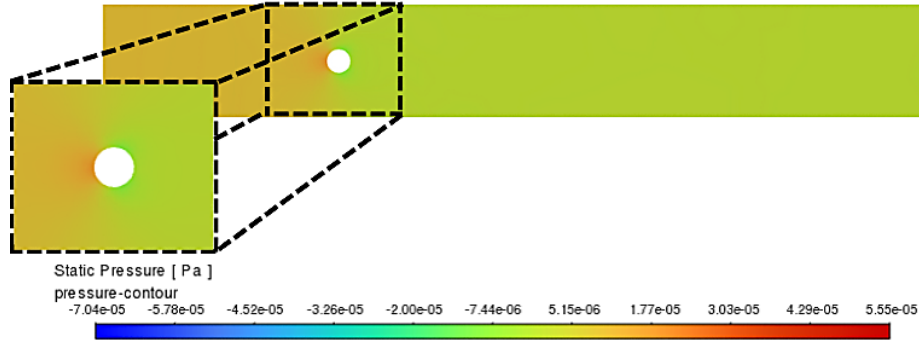


Figure 9: Case A Pressure Contours.

In Case B, the higher velocities produce a stronger pressure drop downstream due to increased dynamic pressure. A localized low-pressure zone appears immediately behind the injection outlet, consistent with Bernoulli's principle and flow acceleration. The pressure gradient along the channel becomes much more pronounced because momentum transport is significantly greater.

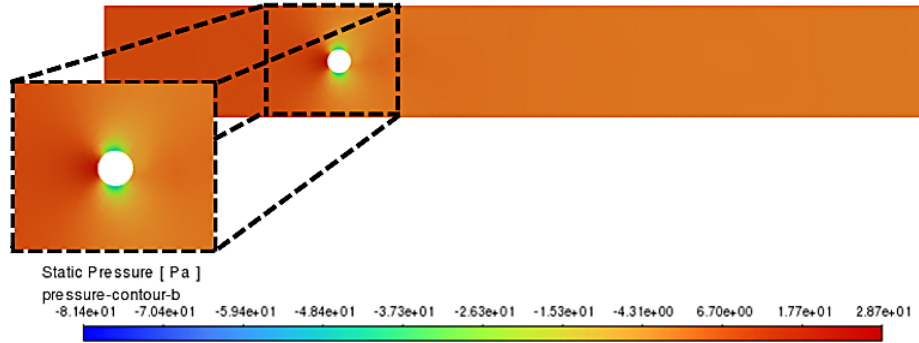


Figure 10: Case B Pressure Contours

3 Analysis of the Flow

Application of Fundamental Principles

To verify that the numerical results produced by ANSYS Fluent were physically valid, two fluid mechanics principles were applied to both Case A and Case B: the continuity equation and the linear momentum equation. These principles provide a framework for evaluating whether the simulated flow behaves consistently with the laws of conservation.

The continuity equation was used to verify mass conservation within the channel. Mass flow rates were obtained at the main inlet, the circular injection inlet, and the outlet, where the inflow was compared to the outflow to determine whether the total mass entering the control volume matched the mass leaving it. A properly converged incompressible simulation should produce a result with negligible mass imbalance, confirming that Fluent is accurately conserving mass across all boundaries.

The momentum equation was evaluated by examining the forces acting on the fluid through pressure and viscous stresses at the channel walls. Fluent integrates the pressure distribution and wall shear stress to compute the net force in the stream direction. Comparing this force with the overall momentum change in the control volume provides a consistency check on the simulated results. A reasonable match indicates that the solver is capturing the correct balance between forces and momentum transport.

Applying these fundamental principles allows the reliability of the numerical solution to be assessed independently of the visual flow patterns. Continuity ensures proper mass conservation, while the momentum balance verifies that the simulated forces correspond to the expected motion of the fluid.

Continuity Analysis Across Cross-Sections

To verify that the numerical solution satisfies mass conservation, the incompressible continuity equation was applied across the main inlet, the circular injection inlet, and the outlet for both Case A and Case B.

For steady, incompressible flow, the control volume form of the continuity equation[1] is:

$$\sum_{CS} \dot{m}_{in} - \sum_{CS} \dot{m}_{out} = 0, \quad (1)$$

where $\dot{m} = \rho VA$ is the mass flow rate, with inflow taken as positive and outflow taken as negative (consistent with Fluent's sign convention). These equations require that the total mass entering the control volume equals the total mass leaving it, and any imbalance should be near zero for a converged incompressible simulation.

Boundary	Case A (kg/s)	Case B (kg/s)
Inlet	9.982	19964
Injection Inlet	0.9237	1231.61
Outlet	-10.90571	-21195.614
Net Error	1.3×10^{-10}	8.1×10^{-4}
Percent Error	$1.2 \times 10^{-9} \%$	$3.8 \times 10^{-6} \%$

Table 2: Mass flow rates at all boundaries for cases A and B.

The mass flow results for both simulations were used to evaluate whether the flow satisfies the incompressible continuity equation. For case A, the sum of the inlet mass flow rates closely matches the outlet flow rate, producing a negligible imbalance in the system. This indicates a near-perfect mass conservation and confirms that the low-velocity case satisfies eq.(1).

For case B, although the velocities and flow rates are significantly larger, the simulation still maintained a very small mass flow rate imbalance. This represents a tiny relative error and is well within the expected numerical tolerance for higher-momentum flow.

Overall, both cases satisfy the continuity equation with high accuracy, demonstrating that eq.(1) is satisfied.

Momentum Analysis Using Pressure Distribution

To evaluate whether the simulated forces are consistent with the flow behavior, the linear momentum equation was applied in the flow direction using the forces obtained from Fluent. In this project, the channel walls (flat wall and circular injection wall) were treated as the primary surfaces exerting forces on the fluid through pressure and viscous shear.

For steady, incompressible flow in the x-direction, the control-volume form of the linear momentum equation [1] can be written as:

$$\sum F_x = \sum_{CS} \dot{m}_{ox} u_{ox} - \sum_{CS} \dot{m}_{ix} u_{ix}, \quad (2)$$

where $\sum F_x$ is the net force acting on the control volume in the flow direction (including pressure and shear forces on the walls), $\dot{m}$ is the mass flow rate, and u_x is the velocity component in the flow direction.

In Fluent, the wall forces in the flow direction are obtained by integrating the pressure distribution and wall shear stress over each wall boundary. The software then reports the separate pressure and viscous contributions, as well as the total force in the specified direction.

Quantity	Case A	Case B
Total Force (N)	9.98×10^{-4}	327.7
Momentum Difference (N)	9.7×10^{-4}	326.7
Net Error (N)	2.8×10^{-5}	1.0
Percent Error	2.9 %	0.31 %

Table 3: Comparison of momentum balance for cases A and B.

The momentum results for both simulations were used to determine whether the flow satisfies the steady x-direction momentum balance in eq.(2).

For Case A, the total force acting on the control volume closely matches the difference in momentum between the inlet and outlet. The error is very small, representing only a few percent difference. This confirms that the momentum balance is satisfied for the low-velocity regime, consistent with the nearly uniform flow behavior observed in the flow visualizations.

For Case B, although the velocities and forces are larger, the numerical force reported by Fluent still agrees with the momentum difference. The error remains less than 1%, which is well within typical numerical tolerance for higher-momentum flows.

Overall, both simulations satisfy the momentum equation with high accuracy, demonstrating that the pressure and shear forces computed by Fluent are dynamically consistent with the momentum transport through the control volume.

4 Conclusion

Through the simulation of discharge flow within a rectangular channel, this project demonstrated how different inlet and injection velocity conditions influence the development of the flow field. By examining streamlines, velocity contours, and pressure distributions, it became clear that increasing the imposed velocities strengthens mixing near the injection region, creates more noticeable pressure gradients, and produces a more defined flow structure throughout the channel.

Continuity checks showed that the numerical solution behaves consistently with mass conservation principles, particularly in the moderate-velocity case, while the momentum analysis revealed how pressure forces on the channel walls increase with higher flow speeds.

Overall, the simulation provided a clear representation of how the flow behaves inside the channel and offered scientific reasoning for the observed patterns, showing how velocity conditions directly affect mixing, pressure variation, and momentum transfer in the system. This project reinforces the connection between theoretical fluid mechanics and numerical modeling, illustrating how computational tools can be used to analyze and interpret complex flow behavior.

References

- [1] Crowe, C. T., Elger, D. F., and Roberson, J. A., *Engineering Fluid Mechanics*, 8th ed., John Wiley & Sons, 2005.
- [2] ANSYS, Inc., *ANSYS Fluent* [Online].
 Available at: <https://www.ansys.com/products/fluids/ansys-fluent>
 Accessed: December 1, 2025.

Appendices

A. Simulation Setup Summary

The simulations were performed using ANSYS Fluent with the following basic settings:

- Solver: steady-state, pressure-based
- Mesh: default mesh
- Initialization: standard initialization
- Convergence: residuals reduced to approximately 10^{-6}
- Boundary conditions: velocity inlets (main and injection), pressure outlet, no-slip walls

This summary applies to both case A and case B, with inlet velocities defined in tables 2 and 3 in the main report.

B. Convergence Plots

B.1 Residuals

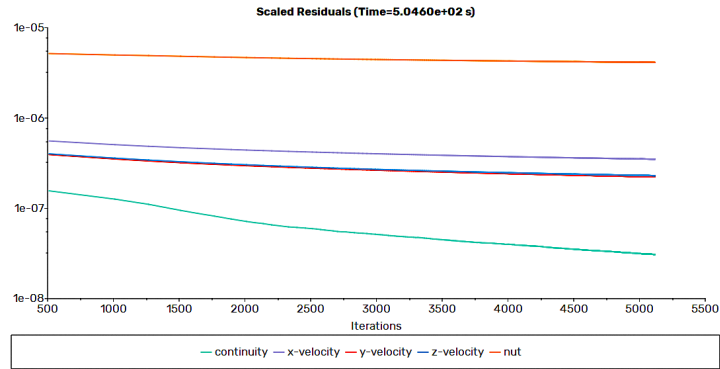


Figure 11: Case A Scaled Residuals

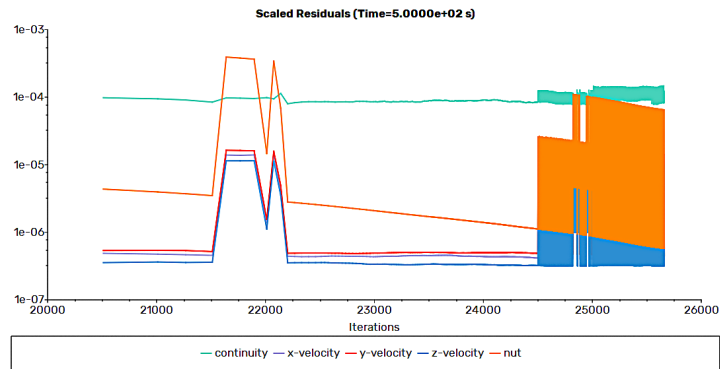


Figure 12: Case B Scaled Residuals

B.2 Drag Coefficient Monitor

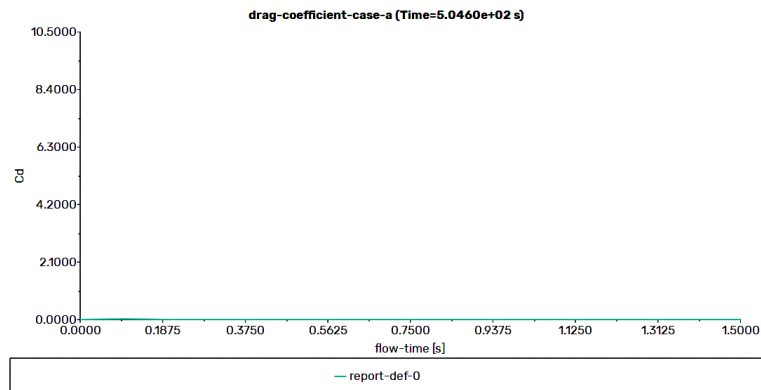


Figure 13: Case A Drag Coefficient

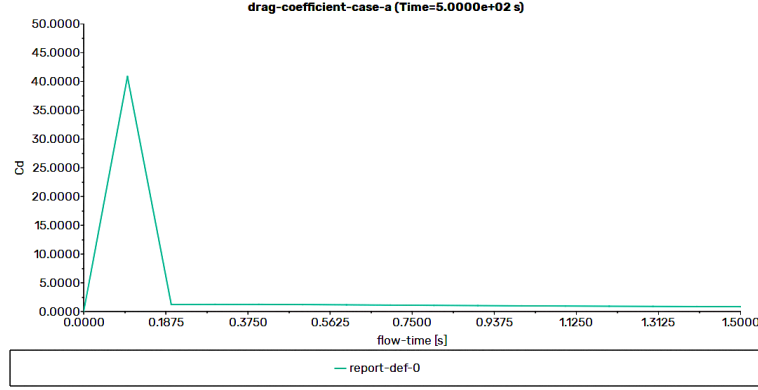


Figure 14: Case B Drag Coefficient

C. Fluent Reports

C.1 Mass Flow and Force Reports

Mass Flow Rate		Mass Flow Rate	
[kg/s]		[kg/s]	
inlet	9.982	inlet	19964
inlet.1	0.92370967	inlet.1	1231.6129
outlet	-10.90571	outlet	-21195.614
Net	1.3243273e-10	Net	-0.00080842236

(a) Case A.

(b) Case B.

Figure 15: Mass flow report for cases A and B.

Forces - Direction Vector (1 0 0)						
Zone	Forces [N]			Coefficients		
	Pressure	Viscous	Total	Pressure	Viscous	Total
wall	2.9944839e-19	0.0007266092	0.0007266092	5.9997674e-22	1.4558389e-06	1.4558389e-06
wall.1	0.00022336707	3.2854202e-05	0.00025622127	4.4753971e-07	6.5826893e-08	5.133666e-07
Net	0.00022336707	0.0007594634	0.00098283047	4.4753971e-07	1.5216658e-06	1.9692055e-06

Figure 16: Force report for case A.

Forces - Direction Vector (1 0 0)						
Zone	Forces [N]			Coefficients		
	Pressure	Viscous	Total	Pressure	Viscous	Total
wall	2.3406663e-13	182.66151	182.66151	4.6897742e-16	0.36598179	0.36598179
wall.1	142.15671	2.9372074	145.09392	0.28482611	0.0058850077	0.29071112
Net	142.15671	185.59872	327.75543	0.28482611	0.3718668	0.65669291

Figure 17: Force report for case B.